

Decoding Customer Departure

An infographic on leveraging Machine Learning to predict, understand, and mitigate customer churn.

The Machine Learning Workflow

Our project follows a comprehensive, multi-stage methodology to build a predictive model that is not only accurate but also interpretable and fair.



1. Data Preprocessing

Cleaning, exploring, and engineering data to build a solid foundation.



2. Model Optimization

Training and tuning multiple models to find the best performer.



3. Comprehensive Evaluation

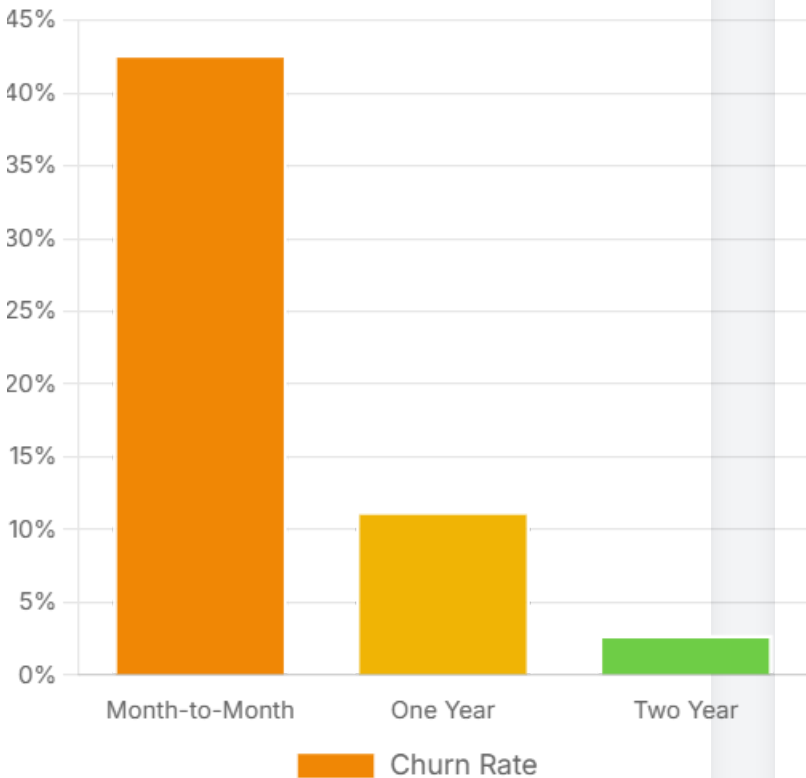
Assessing accuracy, interpretability, and ethical fairness.

Deep Dive into Data Preprocessing

Effective preprocessing is the most critical step. We transformed raw data into high-quality features to fuel our models.

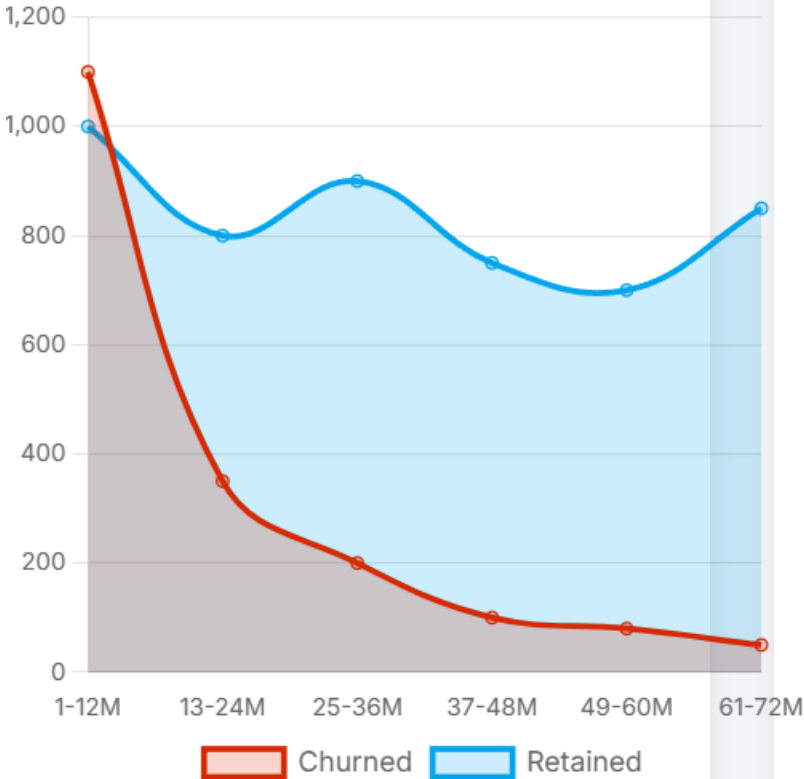
Churn Rate by Contract Type

Our analysis revealed a crucial insight: customers on month-to-month contracts are significantly more likely to churn than those on long-term contracts. This highlights a key area for targeted retention efforts.



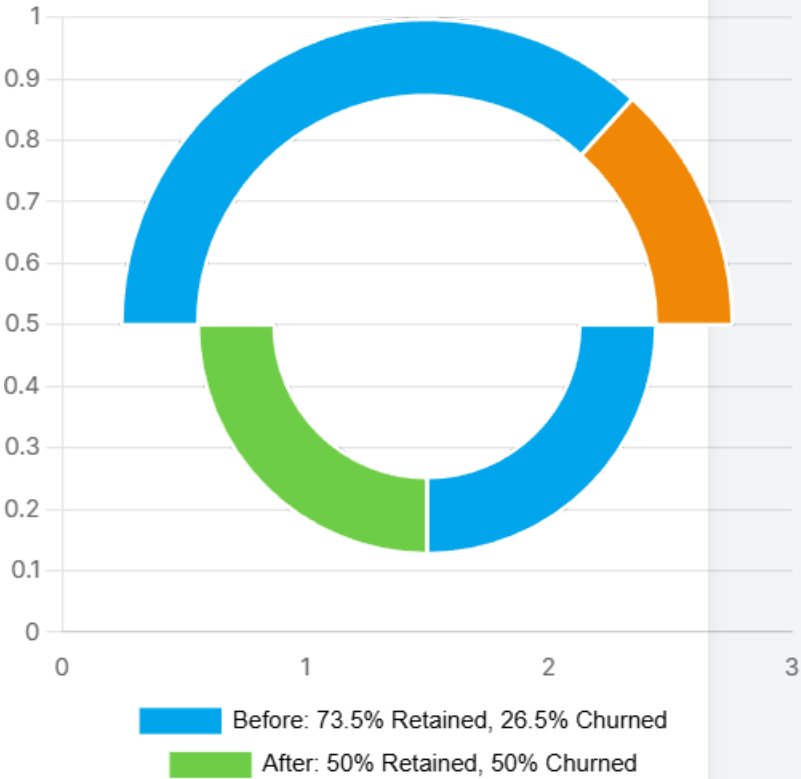
The Impact of Customer Tenure

Newer customers, especially within the first year, represent the highest churn risk. As tenure increases, customer loyalty grows, and the likelihood of churning decreases dramatically.



Balancing the Data with SMOTE

Our initial dataset was imbalanced, with far fewer churners. We used SMOTE to synthetically create new churning samples in the training data, ensuring our model learned to identify them effectively.

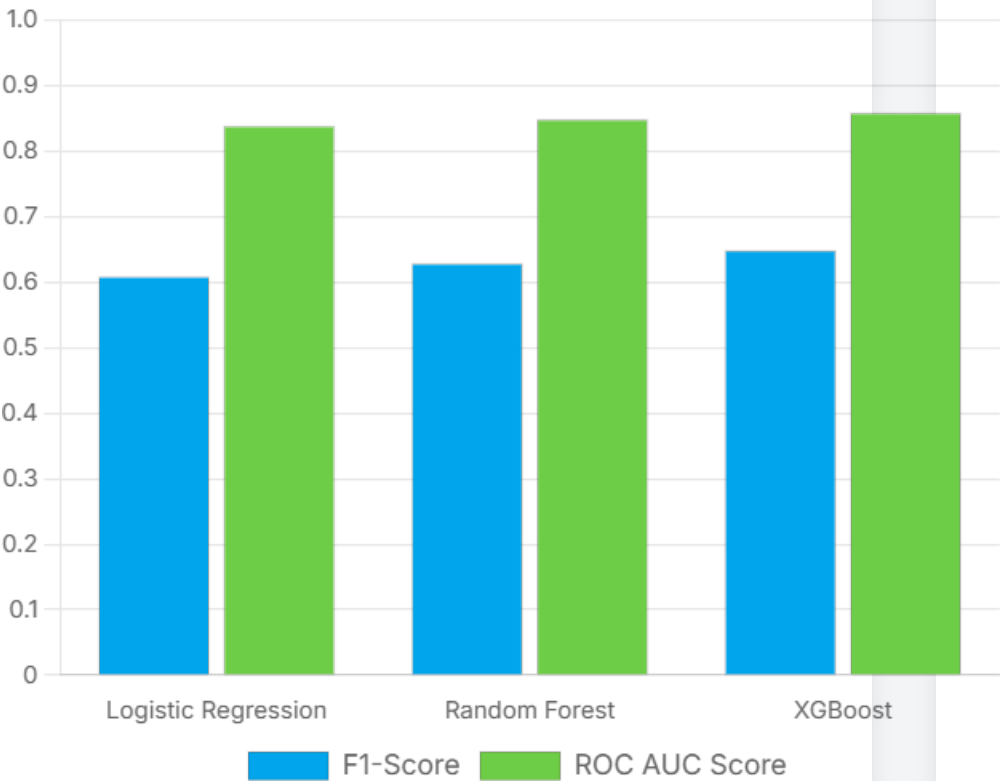


Modeling, Tuning, and Performance

We trained and optimized three distinct models, ultimately identifying a top performer through rigorous evaluation.

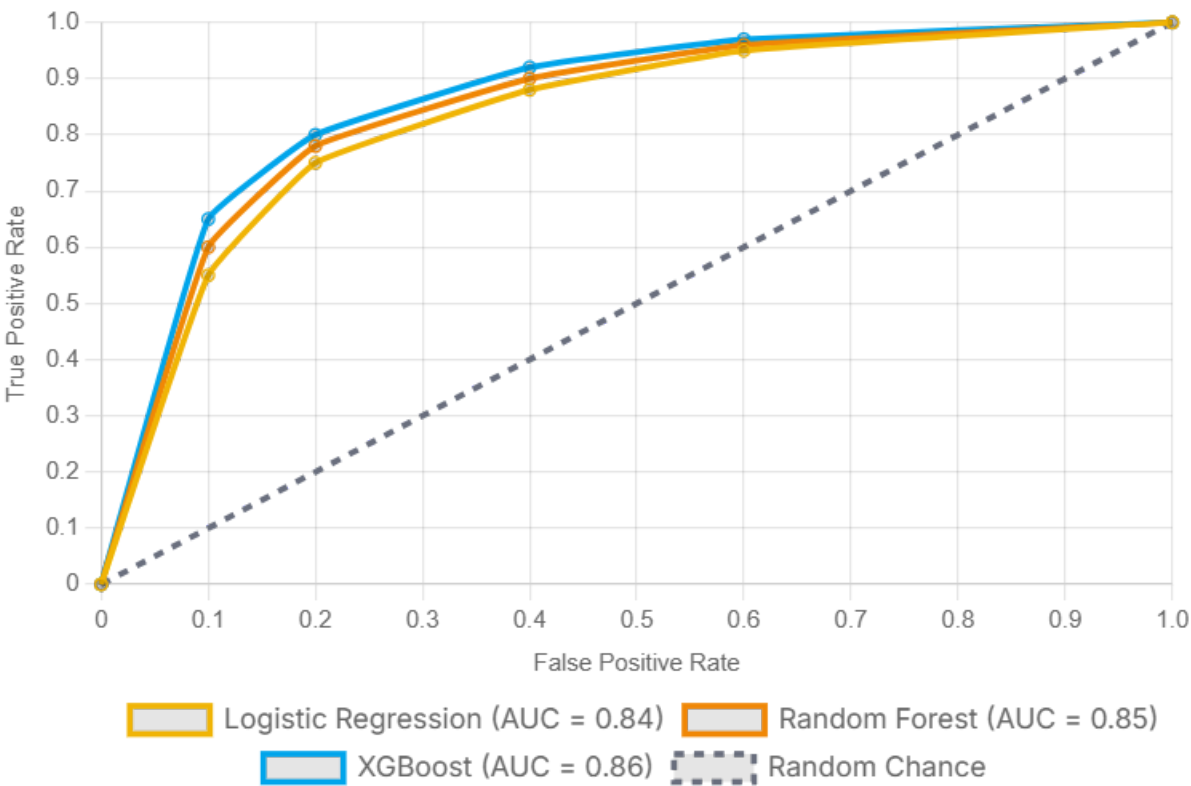
Model Performance Metrics

After hyperparameter tuning, the XGBoost model demonstrated superior performance, particularly in F1-Score and ROC AUC, indicating its excellent balance of precision and recall and its strong ability to distinguish between churners and non-churners.



Model Comparison with ROC Curves

The ROC (Receiver Operating Characteristic) curve visualizes the diagnostic ability of our models. The curve for XGBoost covers the most area (higher AUC), confirming it as the most effective classifier for this task.

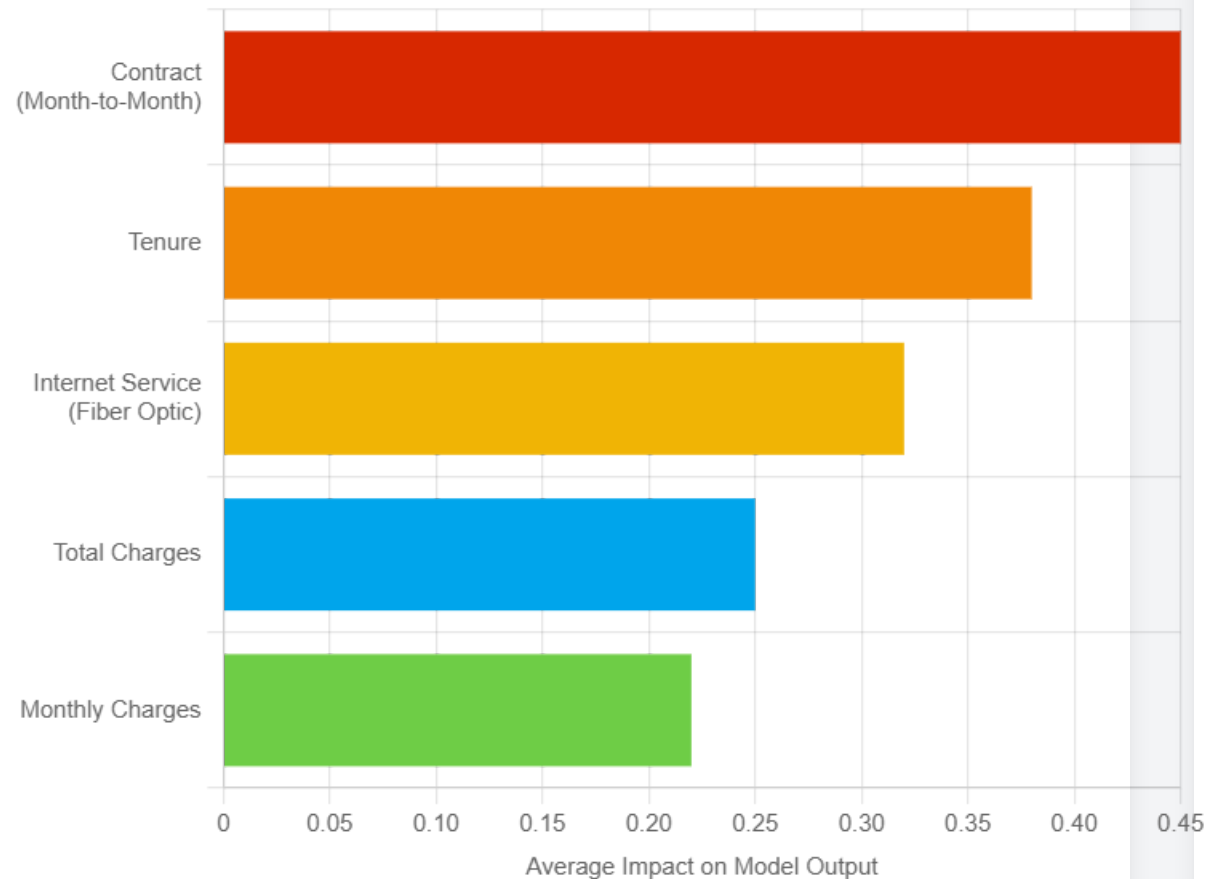


Unlocking Insights & Ensuring Fairness

A great model doesn't just predict—it provides understandable insights and operates ethically.

Top Drivers of Customer Churn

Using SHAP (SHapley Additive exPlanations), we identified the most influential factors driving churn. The model's decisions are heavily influenced by contract type, customer tenure, and whether they have fiber optic internet service.



Algorithmic Fairness Assessment

We critically evaluated our model for bias across gender and senior citizen status. The Demographic Parity Ratio, which measures if churn is predicted at similar rates across groups, is very close to the ideal value of 1.0, indicating no significant bias.

Gender Parity Ratio

0.98

(Close to ideal 1.0)

Senior Citizen Parity Ratio

1.03

(Close to ideal 1.0)

This infographic summarizes the findings of the "Advanced Customer Churn Prediction" project.

Built with HTML, Tailwind CSS, and Chart.js.